The Mayo Clinic

HOSPITAL MANAGEMENT SYSTEM REQUIREMENTS

Client: The Mayo Clinic

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Background

The Mayo Clinic is an American nonprofit academic medical centre based in three major locations, Minnesota, Florida and Arizona. The clinic is currently focused on integrated patient care, education, and research and holds the number 1 rank among hospitals in the United States.

The clinic was in September 1889 and has grown and facilities over the years.

Problem

Due to its growth and the vast increase in the number of patients it has made the management of the hospital an arduous task. Particularly, paperwork and storing of all patients' records was becoming unmanageable.

Business Need

To develop a hospital management software that will manage all hospital operations.

Business Goal/Objective

To develop a software that will store patients records, show availability of beds, manage patients' billing, scheduling a doctor's appointment, and will bring about coordination among the different departments.

List of Stakeholders

Business Stakeholders (Decision Makers)

Stakeholder	Description
Hospital Management	oversee overall hospital operations, monitor performance metrics, generate reports, ensure regulatory compliance, and optimize resource allocation for improved efficiency and patient care
Matrons	oversee nursing staff, ensure compliance with hospital policies, monitor patient care quality, manage ward operations, and track bed occupancy and resource availability
Departmental Heads	manage their respective departments, coordinate staff schedules, monitor department-specific performance, ensure smooth interdepartmental workflows, and oversee compliance with hospital policies and standards
Billing & Finance Department	process patient billing, manage insurance claims, track hospital revenue, generate financial reports, and ensure accurate and timely payments for medical services using the HMS system

Affected Organizational Unit & End Users

Stakeholder	Description
Doctors & Physicians	prescribe treatments, order diagnostic tests, review lab and imaging results, manage appointments, and document patient progress by accessing patients medical records
Nurses & Ward Staff	Manage patient care by accessing medical records, updating treatment details, tracking bed occupancy, and following doctors' instructions for medication and procedures.
Patients & Their Families	Patients and their families use the HMS to access personal medical information, schedule appointments, review billing details, and request or query relevant health records as needed
Front Desk & Administration Staff	Responsible for onboarding, maintaining, and managing patient information while providing necessary details to patients upon request. Including front desk, hospital secretaries, con
Laboratory & Radiology	Responsible for conducting diagnostic tests and imaging procedures for patients before and after treatment

Pharmacy	Ensures patients medication aligns with their specific need and dispenses medication prescribed by doctors
Human Resource	To manage staff records, resources including staff doctor including availability
Finance & Accounting	

Solution Delivery/Project Team

Stakeholder	**Description**
Developers	to design, build, and maintain the system, ensuring seamless integration of features such as patient records, billing, scheduling, and compliance with security and performance requirements
Project Team	oversees the planning, execution, and deployment of the HMS, ensuring alignment with hospital needs. They coordinate between stakeholders, manage timelines, mitigate risks, and ensure successful system implementation and adoption
Business Analysts	ensure the HMS meets business needs, enhance system functionality based on feedback, and support process improvements for efficient hospital management

External Stakeholders

Stakeholder	**Description**
Insurance Companies	Provides funding for patients care and ensures a smooth claim
Medical Suppliers	Provides the hospital with medical equipment, and ensures
Government & Regulatory Bodies	Ensures the hospital follows
Laboratory	receive test requests from doctors, conduct diagnostic tests, and upload test results directly into the system
Legal & Compliance Team	monitor compliance with standards such as HIPAA, manage risk assessments, and oversee legal documentation related to patient care and hospital operations

Proposed System Workflow

Existing Patients

1. Patient visits website where the timings of all the hospital doctors to be displayed on the website.
2. Patients select doctor they would like to visit based on the appointment slot available for that doctor.
3. Patient enters their details and books an appointment.
4. The system displays a booking confirmation screen after booking appointment.
5. A confirmation is sent to patient by email and SMS.
6. System sends out confirmation email and SMS 1 day prior to appointment.
7. On the day of the appointment admin/registration staff check patient in on arrival.
8. System generates a new patient id
9. Patient is checked-in

Doctors

1. Doctor logs into the system.
2. On appointment day or whenever needed the doctor enters patients' id to view patient's record.
3. Doctor orders tests for patient if needed.
4. On day of appointment doctor enters consultation notes within system using patient id.
5. Doctor enters instructions for nurse and ward staff to follow for each patient in the system.
6. Doctor can prescribe medication or further test for client.
7. This will be received by the relevant department (laboratory, pharmacy or radiation) department directly.
8. Laboratory or radiation department performs tests and update patient record.
9. Pharmacy dispensed medication and updates their records accordingly if medication is required.

Nurse and Ward Staff

1. Nurse or ward staff log in
2. While conducting patient care the nurse just looks up the system to understand which medicine or what line of treatment, they need to give the patients.
3. Nurse and ward staff update patient record with treatment accordingly.
4. Doctor can log in to view patient's progress accordingly and discharged

System

1. HMS will total up all the expenses of a patient at one time and produce a complete bill at the end of the consultation or at discharge.
2. At the end of day reports of bed occupancy, doctors' appointments and revenue generated through OPDs is generated for senior management, Total number of OPD patients and admitted patients, which doctors generate the maximum revenue, Total amount of earnings through OPD and admitted patients, Total amount of earnings generated through laboratory and radiology by the HMS system.
3. In case of patients that have insurance, all the insurance details are to be stored in the system for claim processing.

New Patients

1. New patients arrive at the hospital
2. Registration staff registers patient on system
3. Registration staff enters patient insurance details if patient has insurance
4. The system stores all medical records of the patient and their medical history.
5. System generates a new patient id
6. If appointment needed new patient visits website and books appointment as a patient.

Scope

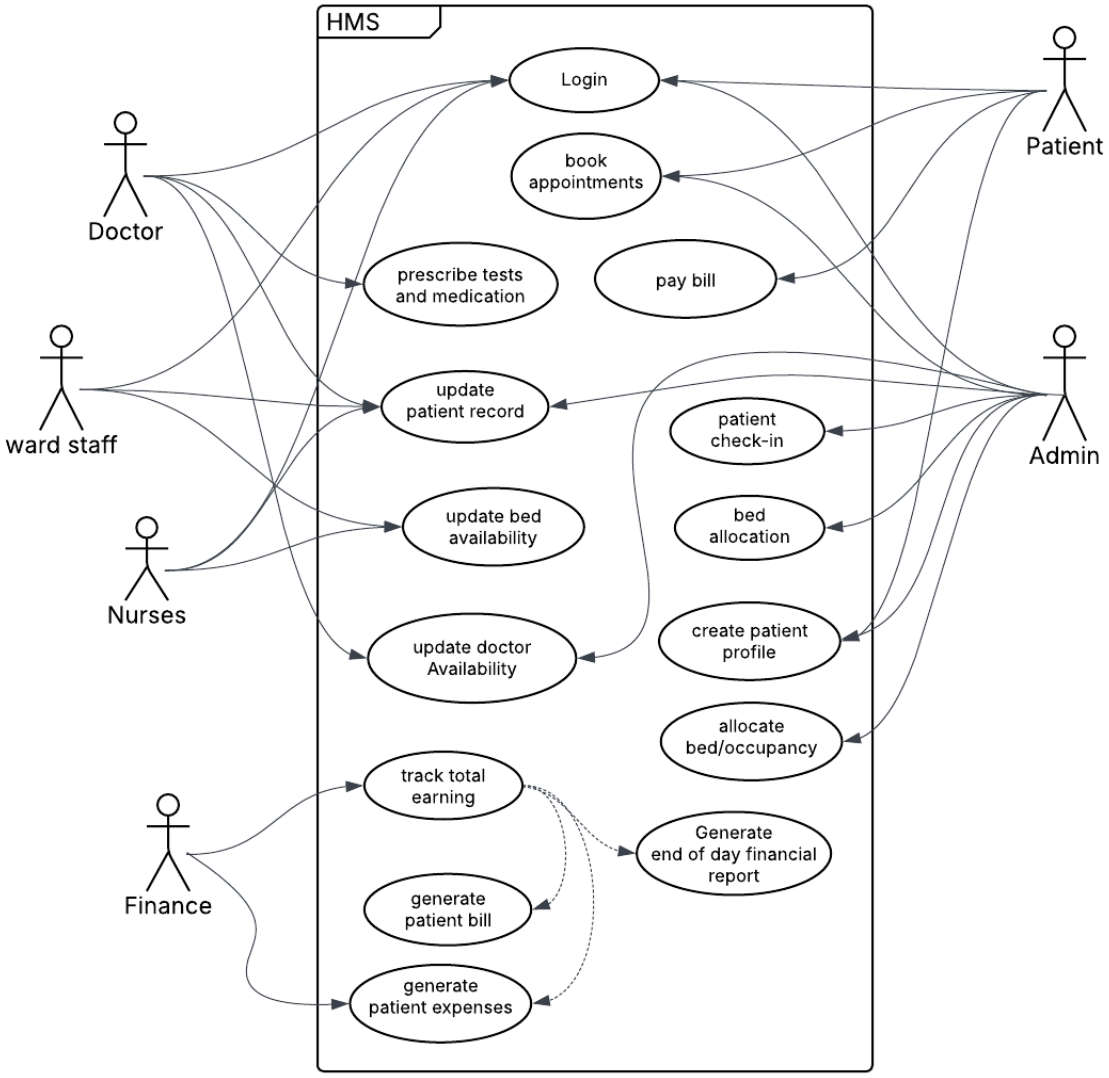
In-Scope

1. Website displays timings of all the hospital doctors available.
2. Can book appointments and write to doctors online calendar/diary
3. Send email and SMS
4. Check patient in to hospital on arrival.
5. System generates a new patient id
6. Display patient ID and details
7. Order test and prescriptions
8. Read and update patient records
9. Batch process all expenses at one time
10. Produce a complete bill at the end of the consultation or at discharge.
11. End of day reports
12. Total amount of earnings
13. Store all insurance details for claim processing.

Out-Scope

1. Automated call to patient
2. Register patient online
3. Register new patients online
4. Book appointment for unregistered patients online
5. Provision for advertising and marketing
6. Manage staff duties
7. Show availability of beds to patients
8. Generate patiently billing and display on website
9. Patient portal or dashboard for patient login

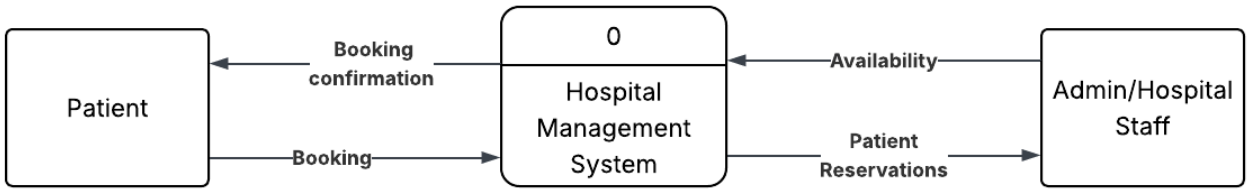
Use Case


HMS
Login
book appointments
prescribe tests and medication
pay bill
update patient record
patient check-in
update bed availability
bed allocation
update doctor Availability
create patient profile
allocate bed/occupancy
track total earning
Generate end of day financial report
generate patient bill
generate patient expenses
Doctor
ward staff
Nurses
Finance
Patient
Admin

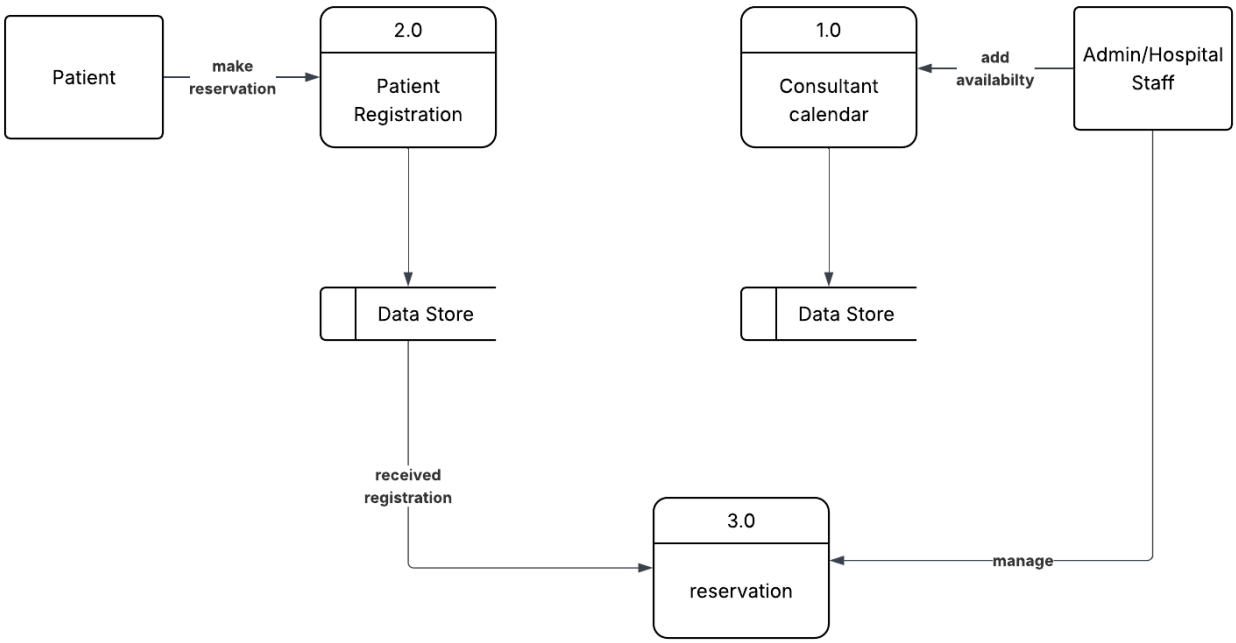
Main Features of HMS

1. Update staff availability
2. timings of all the hospital doctors displayed on the website.
3. Booking of appointments for patients with a doctor
4. confirmation email and SMS to patients 1 day prior to appointment.
5. Patient check-in by staff
6. Patient ID generation
7. Bed and facility allocation
8. General staff login.
9. patient profile creation
10. updating patient record
11. Prescribing tests and medication
12. Total up all patient expenses
13. patient billing
14. Generating end of day reports for management
15. Tracking total earnings

Data Flow Diagram for HMS



Context Diagram



Level 0 diagram

Requirements

Functional Requirements

Patient Management

1. The system should allow registration of new patients
2. The system must generate unique IDs for each new registered patient
3. The system must store and retrieve a patient's medical history.
4. The system should delete patient records upon discharge, if required.

Appointment Scheduling

1 The system must display doctor availability by time slot.
2 The system should allow patients to book appointments online.
3 The system should send appointment reminders via SMS and email 24 hours before the appointment.

Bed Occupancy Tracking

1. The system should track all hospital beds
2. The should update their occupancy status every 6 hours.
3. The system should generate daily reports of bed occupancy.

Billing

1 The system must generate consolidated bills for patients at discharge.
2 The system should support billing breakdown by department or service.

Laboratory & Diagnostics

1 The system must allow doctors to request tests and scans electronically.
2 The system should notify the lab/radiology department of ordered tests.
3 The system should upload and store lab reports accessible via patient ID.

Staff and Schedule Management

1 The system could store the schedules of nurses and ward staff.
2 The system should allow doctors and nurses to access treatment instructions for patients.

Communication

1 The system should allow hospital staff to update patient records in real time.
2 The system could allow patients to request information via the portal.

Insurance & Compliance

1 The system must store and manage patient insurance details.
2 The system should support insurance claim processing workflows.

Reporting

1 The system should generate reports on doctor appointments
2 The system should generate reports on revenue by department
3 The system should generate reports on could bed occupancy trends

Non-Functional Requirements

Performance

1. The system shall respond to all user actions under **1 second**.
2. The system must support at least **500 concurrent users** without affecting performance.

Availability

1 The system shall be available **24/7**, with minimal downtime (with less than 1% of the time).
2 Scheduled maintenance must be communicated in advance
3 Scheduled maintenance should only occur during low-usage hours.

Security

1 The system shall store all data securely using encryption
2 Access to patient data should be role-based
3 Audit trails must be stored for every data access and modification.
4 The system must comply with **data privacy and GDPR regulations)**.

Usability

1 All screens and user interfaces should be **intuitive, self-explanatory**, and require minimal training.
2 The interface should use **consistent labelling, input formats, and navigation elements** across modules to assist in usability.

Data Integrity & Backup

1 The system must perform **automatic daily backups**.
2 The system must ensure that no patient or transaction data is lost due to power failure or system crash.

Maintainability

1 The system must log all application and system-level errors for troubleshooting.

Scalability

1 The system must be scalable to accommodate future expansion, such as additional hospital branches or new services.
2 The database must support data growth for at least the next **5 years** without performance degradation.

Business Requirements

1. Improve Operational Efficiency

- The HMS must streamline hospital operations by reducing manual paperwork and documentation.
- Automate appointment scheduling, billing, and inter-departmental communication to reduce administrative burden.

2. Centralized Patient Information

- Provide a single, secure system for storing and retrieving all patient medical records and history.
- Ensure ease of access to real-time patient data for authorized personnel.

3. Enhance Patient Experience

- Allow patients to book appointments online and receive automated reminders.
- Provide access to their medical records, billing details, and test results via a secure portal.

4. Optimize Resource Utilization

- Track and manage hospital beds, staff schedules, and test facilities to ensure optimal usage.
- Generate reports to identify bottlenecks and underutilized resources.

5. Support Informed Decision-Making

- Generate accurate and timely reports for hospital management to support planning and revenue analysis.
- Include KPIs such as daily bed occupancy, doctor performance, department revenue, and patient traffic.

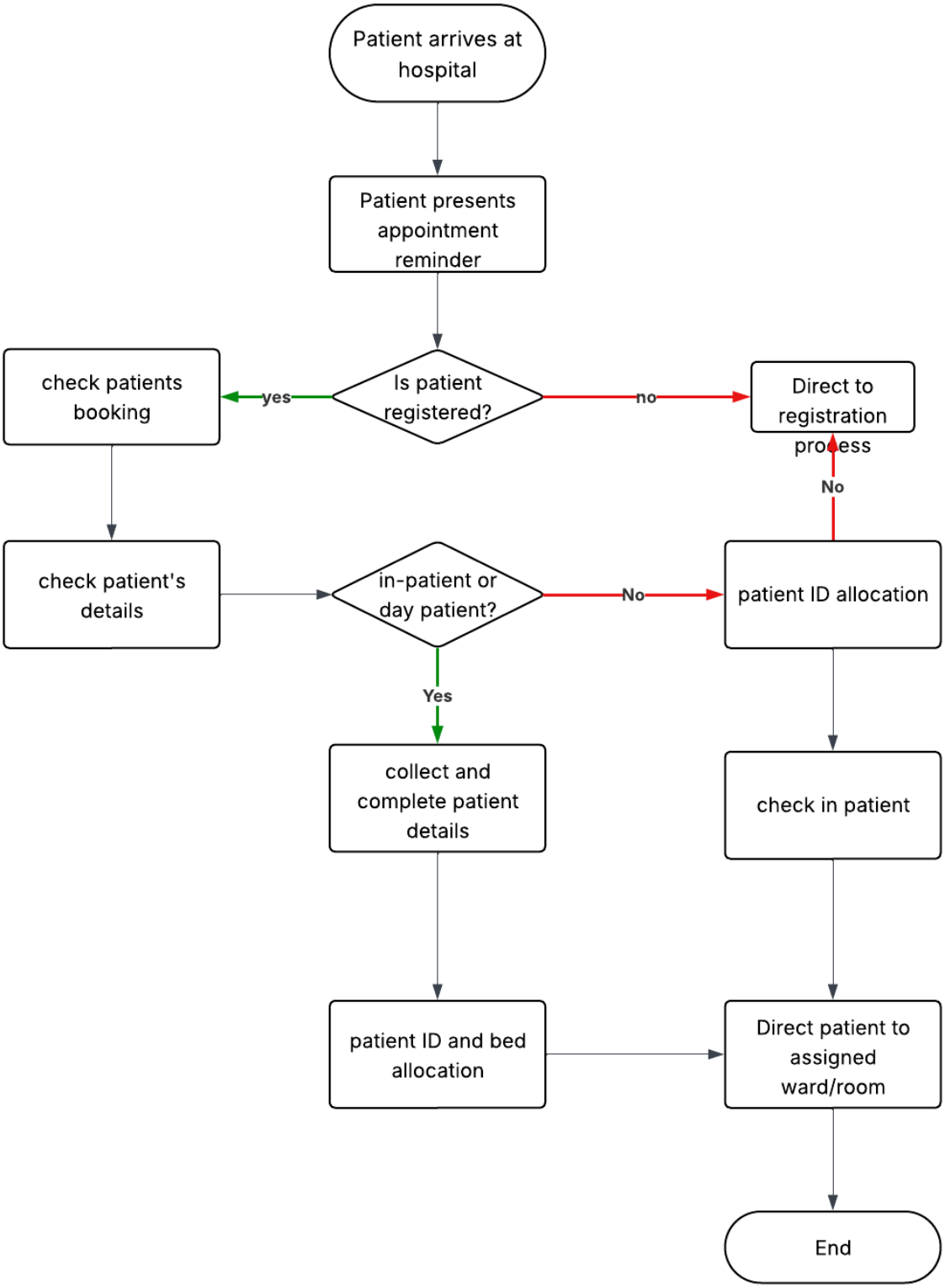
6. Enable Insurance and Financial Integration

- Maintain insurance details for each patient and support claim processing.
- Integrate billing data across departments for consolidated and accurate financial reporting.

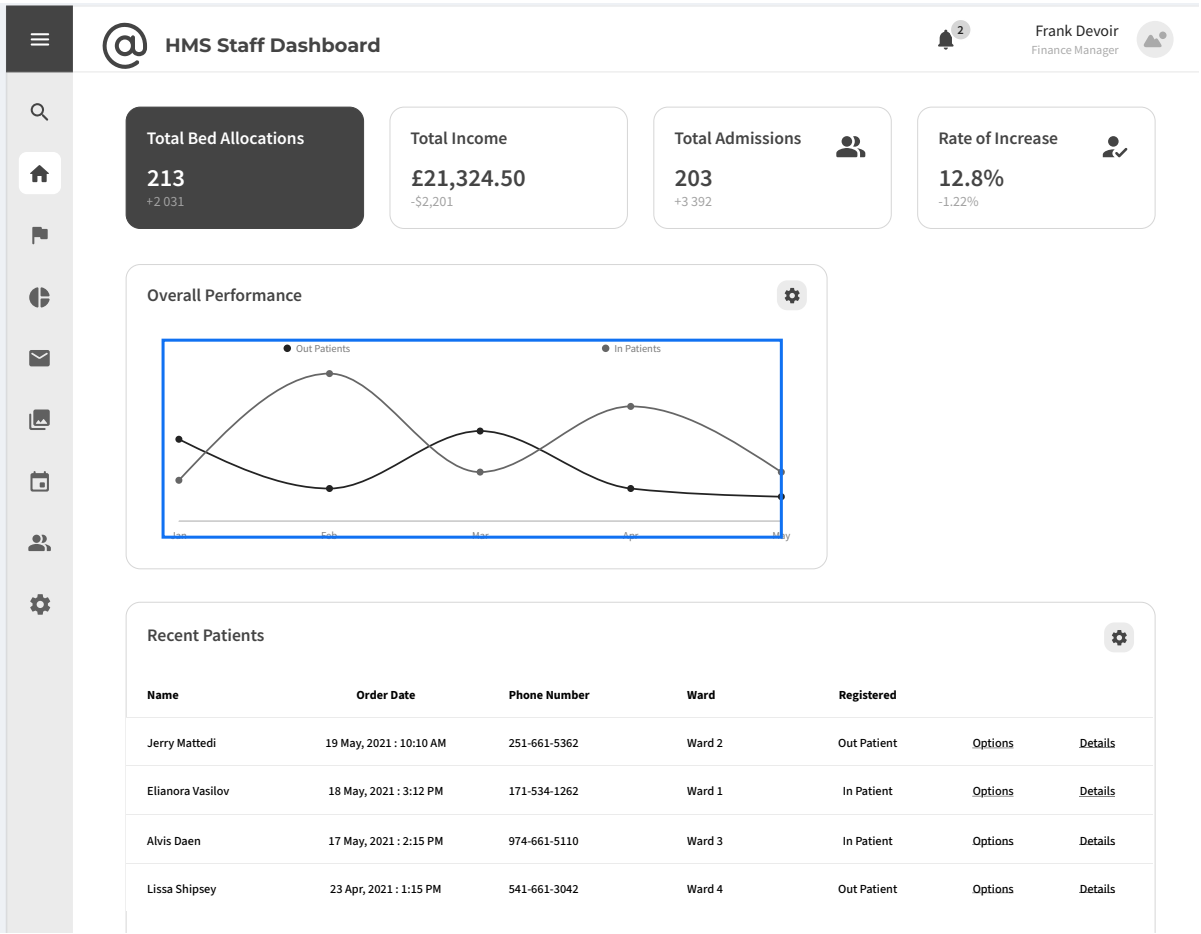
7. Ensure Regulatory Compliance and Data Security

- Maintain confidentiality and integrity of patient information in accordance with healthcare regulations (e.g., HIPAA).
- Implement robust access control and audit trails.

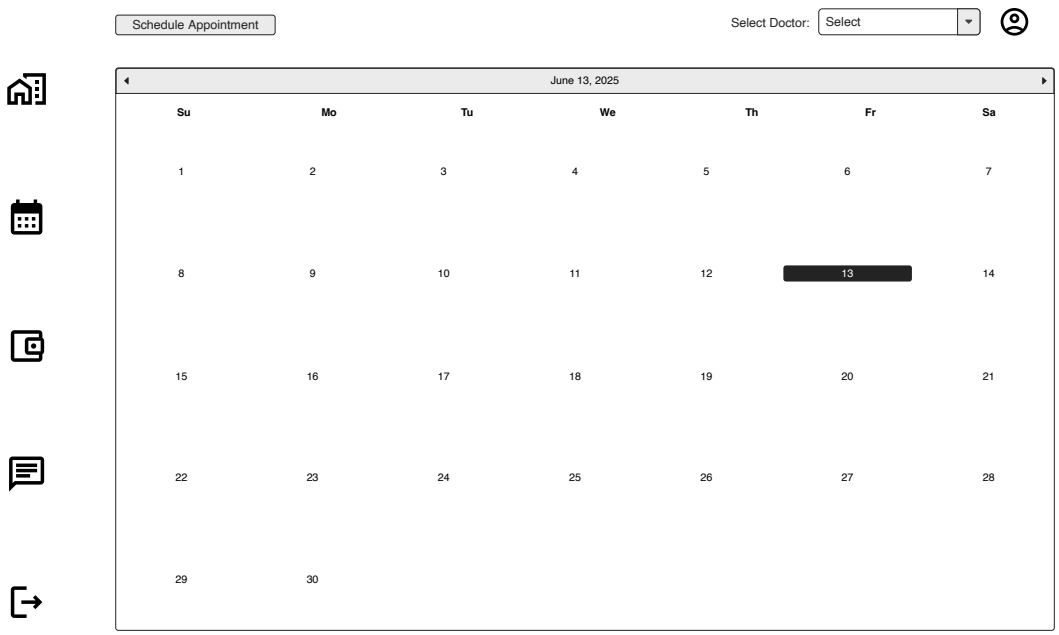
Flowchart for Patient's Admission Process



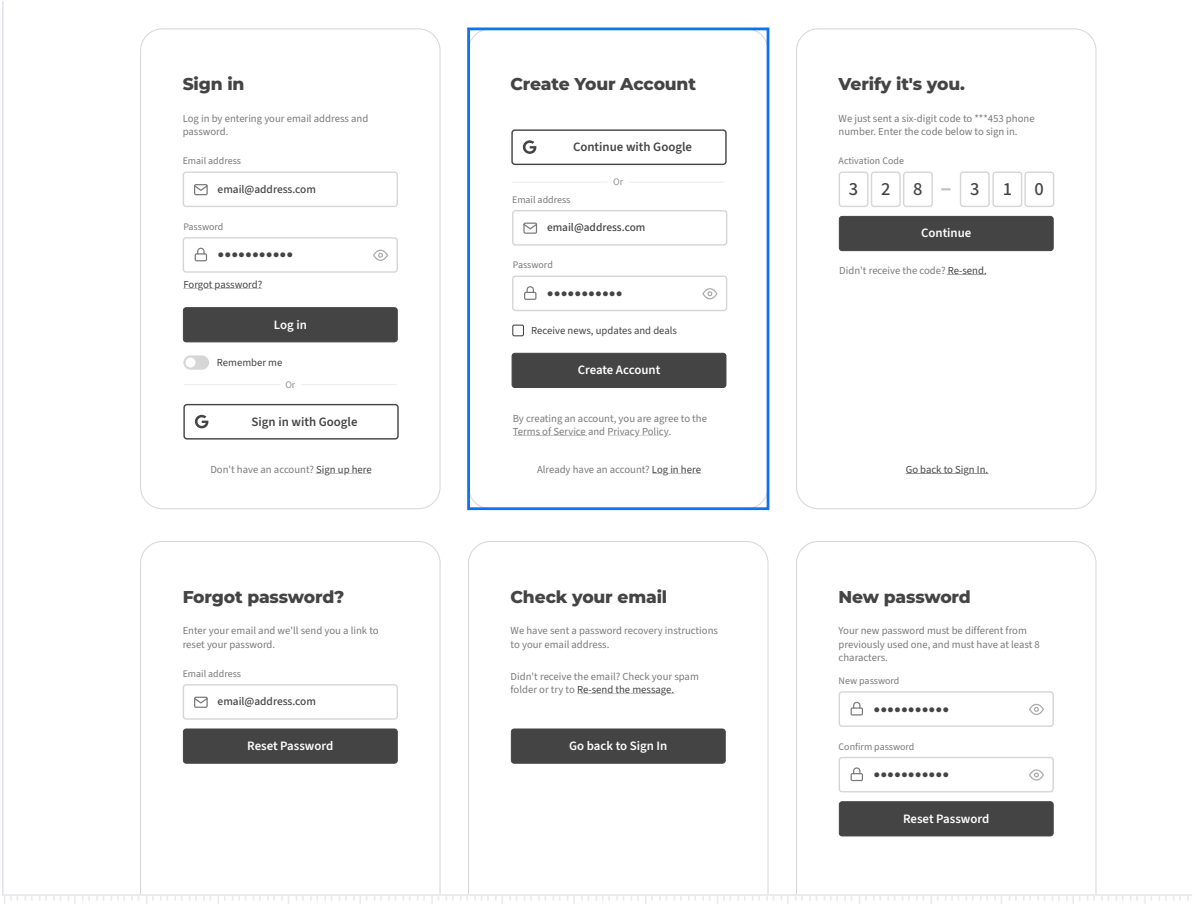
Mock Screens



Financial Report Dashboard 1.0



Appointment Scheduling



Sign up/Sign in

Current state

The hospital uses a system called the Hospital Management System which is designed to manage all hospital operations.

HMS is designed to store patients records, show availability of beds, manage patients' billing, scheduling a doctor's appointment

Disadvantages

Future state

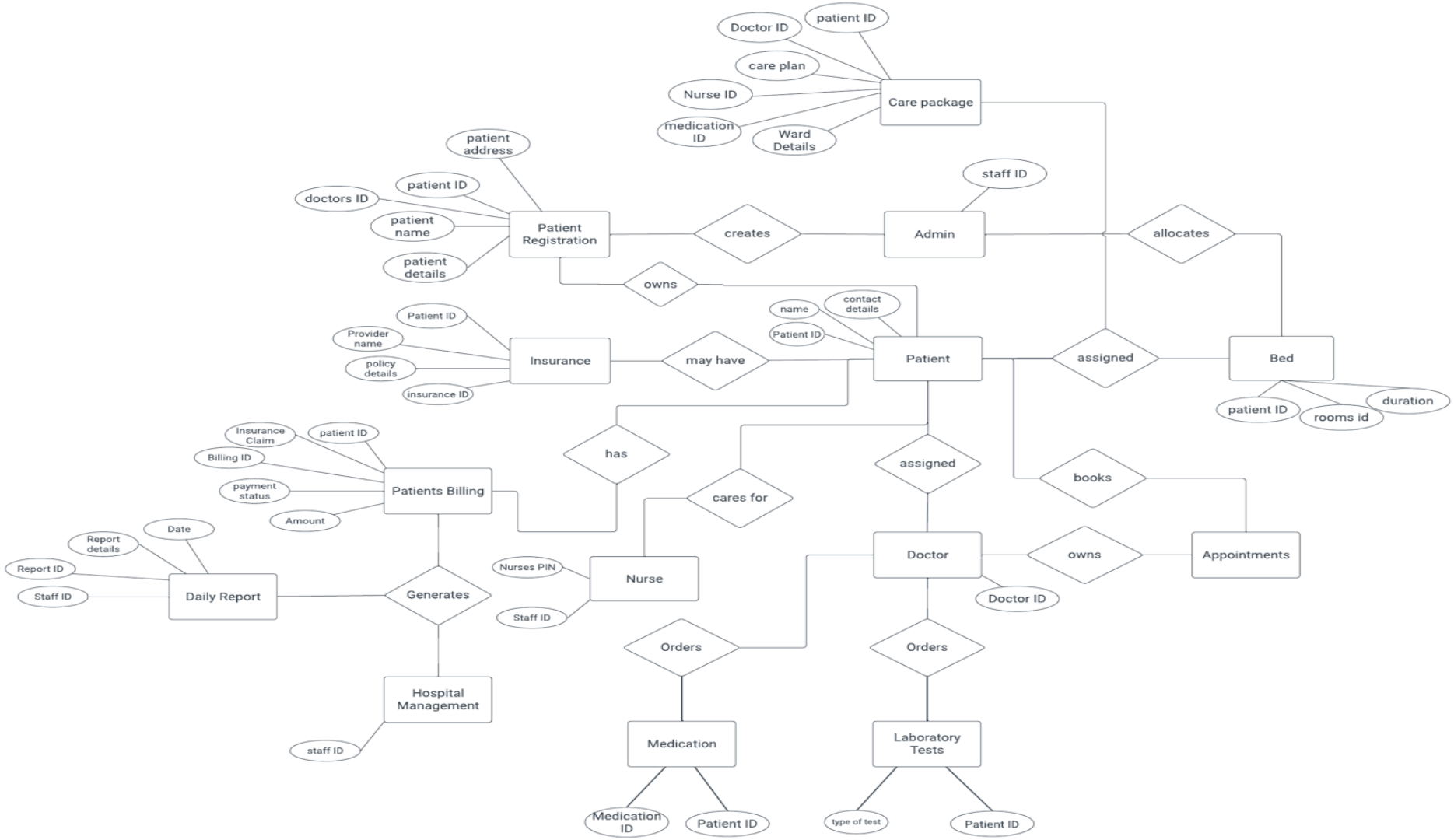
1. will bring about coordination among the different departments.
2. The timings of all the hospital doctors to be displayed on the website. Patients can select the doctor they would like to visit based on the appointment slot available for that doctor. The system shall book the appointment for that patient with the doctor selected.
3. System shall send email and SMS reminders to the patient one day prior to the appointment date to remind the patient of the appointment.
4. When a new patient comes to the hospital, they are registered in the system. The system can store all medical records of the patient and their medical history. The registration staff will give each patient a patient ID. This ID will be used by the patient throughout their stay in hospital. The patient ID will be deleted from the system when the patient checks out.
5. HMS will keep tracks of all the beds in the hospital. It will show the list of all the occupied and unoccupied beds in the hospital. Every 6 hours, the person in charge for the hospital floor will update the bed occupancy.
6. HMS will total up all the expenses of a patient at one time and produce a complete bill at the end of the consultation or at discharge. This will save time and effort for each department as they need not produce separate bills for a single patient.
7. If a doctor needs to prescribe any tests for any patient, he will enter the same in HMS. This will be received by the laboratory or radiation department directly. The patient's tests are done, and the reports are uploaded by the department in the HMS. The doctor can log in the system and enter the patient ID and view them directly. This eliminates the need of paper reports and retrieval is also easy.
8. Reports are generated from the HMS for senior management to have a clear understanding on the hospital's revenue, expenses, bed occupancy, and other details.
    a. Bed occupancy for each day
    b. Doctors' appointments and revenue generated through OPDs.
    c. Total number of OPD patients and admitted patients
    d. Which doctors generate the maximum revenue
    e. Total amount of earnings through OPD and admitted patients
    f. Total amount of earnings generated through laboratory and radiology

9. It stores the names and timings of the nurses and ward boys on duty with their respective ward numbers.
10. All the instructions given by the doctors for the nurse to follow for each patient are entered in the system. While conducting patient care the nurse just looks up the system to understand which medicine or what line of treatment, they need to give the patients.
11. In case of patients that have insurance, all the insurance details are to be stored in the system for claim processing.

Advantages of HMS:

- Reduce operating costs of the hospital
- Provide reports to senior management for better decision-making
- Saves patients' time
- Keeps patients' medical records secure and stored in cloud
- Keeps track of empty and filled beds in the hospital
- Easy access to patient data
- Reduces documentation in the hospital

DFD Diagram


Doctor ID
patient ID
care plan
Nurse ID
Care package
medication ID
Ward Details
patient address
staff ID
patient ID
doctors ID
patient name
Patient Registration
creates
Admin
allocates
patient details
owns
name
contact details
Patient ID
Patient ID
Provider name
Insurance
may have
Patient
assigned
Bed
policy details
insurance ID
duration
patient ID
rooms id
Insurance Claim
patient ID
Billing ID
has
assigned
books
payment status
Patients Billing
cares for
Amount
Date
Report details
Doctor
owns
Appointments
Report ID
Nurses PIN
Nurse
Staff ID
Daily Report
Generates
Doctor ID
Staff ID
Orders
Orders
Hospital Management
Medication
Laboratory Tests
staff ID
Medication ID
Patient ID
type of test
Patient ID

Patient: Patient ID (PK), Name, Date of Birth, Address, Contact Info, Insurance Info, Medical History

Doctor: Doctor ID (visualize Specialization, Availability Schedule, Department

Appointment: Appointment ID (PK), Patient ID (FK), Doctor ID (FK), Date, Time, Status

Bed: Bed ID (PK), Ward Number, Room Number, Occupancy Status

Billing: Bill ID (PK), Patient ID (FK), Total Amount, Payment Status, Insurance Claim ID (FK)

Insurance: Insurance ID (PK), Patient ID (FK), Provider Name, Policy Number, Coverage Details

Laboratory Test: Test ID (PK), Patient ID (FK), Test Type, Test Date, Test Result

Radiology Test: Radiology ID (PK), Patient ID (FK), Test Type, Test Date, Test Result

Staff (Nurses/Ward Staff): Staff ID (PK), Name, Role, Shift Timing, Ward Assignment

Hospital Management: Management ID (PK), Name, Role, Department

Instruction: Instruction ID (PK), Patient ID (FK), Doctor ID (FK), Nurse ID (FK), Treatment Plan, Medication Details

Report: Report ID (PK), Report Type, Generated Date, Content Details

Relationships

Relationship Entities Involved Description

Patient books Appointment Patient ↔ Appointment One patient can have multiple appointments.

Doctor attends Appointment Doctor ↔ Appointment One doctor can have many appointments.

Patient occupies Bed Patient ↔ Bed One patient can occupy one bed at a time.

Patient has Billing Patient ↔ Billing One patient has one or more billing records.

Patient has Insurance Patient ↔ Insurance One patient may have one insurance record.

Doctor prescribes Laboratory/Radiology Test Doctor ↔ Laboratory Test / Radiology Test A doctor can prescribe many tests.

Staff follows Instructions Staff ↔ Instruction Staff members follow multiple patient instructions.

Hospital Management generates Reports Management ↔ Report Management generates multiple reports.